

CS 217: Practice questions on Generative Classifiers

1. TRUE/FALSE: Whenever a set S of labeled instances is split into two sets S_1 and S_2 , the average entropy will always decrease, irrespective of the split attribute or the split point.

Give reasons if TRUE, or a contradictory example if FALSE.

2. The following dataset will be used to learn a decision tree for predicting whether a person is Happy (H) or Sad (S) based on the color of their shoes, whether they wear a wig and the number of ears they have.

Color	Wig	Num. Ears	Emotion
G	Y	2	S
G	N	2	S
G	Y	2	S
B	N	2	S
B	N	2	H
R	N	2	H
R	N	2	H
R	N	2	H
R	Y	3	H

- (a) What is Entropy(Emotion | Wig=Y)? ..1

$$-\frac{1}{3} * \log(\frac{1}{3}) + -\frac{2}{3} * \log(\frac{2}{3}) = \frac{1}{3} * \log(3) + \frac{2}{3} * \log(\frac{3}{2}) = 0.918$$
- (b) Which attribute would the decision-tree building algorithm choose to use for the root of the tree (assume no pruning). ..1
 Information gains on various attributes are as follows:
 $IG(Color) = 0.76885$
 $IG(Wig) = 0.07278$
 $IG(Ears) = 0.10219$
 Hence, assuming entropy based information gain to be the goodness of an attribute, we would use color for a split at the root of the tree.
- (c) Draw the full decision tree that would be learned for this data (assume no pruning). ..1
 The first split is on color. For branch corresponding to color B, neither wig nor number of ears offers any further information gain. Hence, we can assign a label of 0 or 1 to the leaf at the end of the branch for color B, and stop building the tree.
- (d) What would be the training set error for this dataset? Express your answer as the percentage of records that would be misclassified. ..1
 One example in the training set would be misclassified (either 4th or 5th example in the dataset depending on the tree), leading to a train error of 11.11 percent.

3. Suppose we are trying to construct a decision tree on data with only numerical attributes and binary class labels $\{1,2\}$. Assume one attribute x_1 and let us denote the data as: $(x_1^1, y^1), \dots, (x_1^N, y^N)$. We evaluate the error of a split $x_1 \leq v$ as follows. Let S_1, S_2 denote the left and right subsets of the data. For each set, all examples part of the majority

class label in the set are correctly classifier and incur no error. For the minority class examples, the error is the distance of the attribute from v . For example, consider the dataset $(0, 1), (1, 2), (2, 1), (4, 2), (5, 2)$ and the split condition $x_1 \leq 0.5$. This creates $S_1 = (0, 1)$ and $S_2 = \{(1, 2), (2, 1), (4, 2), (5, 2)\}$. Set S_1 has zero error. In set S_2 , the minority example $(2, 1)$ incurs an error of $2-0.5=1.5$.

- (a) What is the value of v in the split condition $x_1 \leq v$ for which the split error is minimized for the above example? ..1

$x_1 \leq 2$. Error=1.

- (b) Why do you think this might be a better criteria for splitting on numerical columns than information gain or Gini which only consider a 0/1 notion of error? ..1

If leads to choice of split conditions that are sensitive to how far the misclassified examples are from the margin.

- (c) Show an example dataset where the best split based on entropy is different from one based on the above criteria. ..2

Choose one negative at 1000, two positives at 1,1, two negatives at -1, -1,. Entropy will choose v between -1 and 1 whereas the above error function will pick a v between 1000 and 1.